

The Triadic Stress Index in Financial Markets

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Abstract

Recent surveys of soil microbial networks under contrasting management report a consistent picture: agricultural inputs raise modularity and shift network organisation toward compartmentalised configurations that their authors read as less mutualistic and more fragmented [3]. The same literature proposes network properties themselves, rather than species counts, as biomarkers of soil condition [2]. What it does not supply is a mathematical quantity for the thing being described, which is the emergence of **structural monopoly**: a few nodes accumulating the interactions the rest lose. The accompanying theoretical work constructs that quantity, from topological entropy, Wasserstein curvature bounds and the non-equilibrium thermodynamics of complex networks [1]. This paper takes that framework, without alteration, to the correlation network of financial assets, and asks what it says about systemic stress.

We call the resulting quantity the **Triadic Stress Index**, TSI. The contribution is a conjunction rather than a single number: node-level attribution included with no parameter to get wrong, detection on a par with the best spectral measure, the cleanest alarms of anything tested, and a different underlying primitive. The rest of this abstract develops each in that order.

TSI is built from the number of closed triangles in the weighted network, together with density, modularity and degree variance. Its orientation is reversed for this domain, because the configuration that means health in a soil community means crisis in a market. We test it across **five markets**, equities including banking crises and the AI sector, cryptocurrencies, commodities, foreign exchange and sovereign debt, spanning 2006–2026, against labelled crisis episodes, at a fixed alarm budget, out of sample, with block-bootstrap intervals.

What is different about it comes first. The index carries a per-node decomposition, $\text{diag}(A^3)$, that names which asset is carrying the concentration, and it has no parameter to select. Spectral attribution must choose how many components to read: with the right rule it matches the triangle count, with a fixed $k = 2$ it loses a quarter of its accuracy when that value is wrong, and with the equally standard Kaiser rule it collapses to zero. Against the only published per-node alternative on correlation networks, the local balance index, the triangle decomposition scores 0.97–0.99 against 0.33–0.84, and the ordering survives a benchmark rebuilt to favour the rival. We also report, because it is true and useful, that on real matrices the far simpler node degree gives the same answer.

On detection it matches the best available measure. Under a like-for-like protocol, TSI ties the effective rank and the Vendi score, the sharpest spectral baselines we could find, and does so across two samples and three independent crisis definitions. It beats the Absorption Ratio, the industry standard used by MSCI and central banks, by a wide margin (F1 gap 0.273 out of sample, 95% CI [0.10, 0.39], $p < 0.0005$), although that margin narrows under the strictest labelling. Its alarms are the cleanest of anything tested, 4.0% unexplained against 14.7% and roughly 59%, and it shares only half of them with its nearest relative, so it is reading something adjacent rather than identical.

We are explicit about register. A lead-lag analysis shows the cross-correlation with the stress index peaking at zero lag and decaying symmetrically: this is a coincident state index, not a forecast, and every claim here is framed accordingly. Two families of event fall

outside it by construction, single-asset divergence and slow geopolitical compounding, and we close the obvious remedy for the first with a null: signing the correlation leaves the triangle count unchanged, because positive semi-definiteness bounds how frustrated a triangle can be.

Two families of market event fall outside what the index is built to detect, by construction rather than by error: single-asset divergence and slow geopolitical compounding.

Code and data: <https://github.com/BiomeMakers/TSI-OmegaS>

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1. Introduction

TSI began as a descriptor of microbial co-occurrence networks in agricultural soils, where resilience indices were derived from network transitivity and competitive exclusion [4], and where the same four quantities, clustering, density, modularity and the proportion of exclusion edges, were observed to move together along a management gradient. The accompanying theoretical work [1] formalises that construction and establishes that its central quantity, the trace of the cube of the adjacency matrix, proportional to the number of closed triangles in the graph, is domain-agnostic: any system representable as a weighted network admits the same computation.

This motivates a natural question. If the four factors that separate a managed soil community from an undisturbed one are computed instead on the correlation network among financial assets, do they register the kind of structural concentration that characterises a system under systemic stress, the configuration Minsky described as fragility built up in plain sight [11]? We call the resulting quantity the **Triadic Stress Index**, TSI, and write it that way throughout, in prose and in equations.

A note on which factor is active, because that does not travel between constructions and it is the first thing a reader should check. The theoretical work [1] reports a direct measurement in a setting where the adjacency matrix is obtained by passing weight correlations through a bounded sigmoidal map. There the clustering factor C is driven against a constant and is numerically inert, and the operative factor is the degree-variance term Coex . That is a property of how A is built, not of the index. Here A is a correlation matrix among assets, built without any saturating nonlinearity, so C is free to vary and contributes to TSI as defined; we checked this rather than assumed it, and the excess ratio C/D departs from unity on these matrices, whereas under a sigmoidal construction it equals unity to four decimal places. The two settings are therefore consistent: the same four factors, with the clustering channel inert in one construction and active in the other.

We are deliberate about the verb. This paper tests whether TSI *registers* such a state, not whether it *forecasts* its onset. Section 6.7 reports a lead-lag analysis showing the distinction matters: TSI is a coincident state index, not a leading indicator, and we frame every claim accordingly.

Three claims are made and each is stated so that it can fail. **H1, detection:** the index separates labelled crisis windows from calm ones at least as well as the sharpest spectral measures available, at an equal alarm budget. It fails if a baseline beats it with an interval excluding zero. **H2, attribution:** the per-node decomposition names the asset carrying the concentration without any parameter to select. It fails if a rule with a parameter beats it, or if a published per-node alternative does. **H3, register:** the index is coincident with stress rather than leading it. It fails if the cross-correlation with an external stress measure peaks at a negative lag.

The order of what follows is deliberate: first what the index does that the alternatives do not, then where it merely equals them, then the boundary of what it can be asked to do at all. Results that did not survive testing are reported alongside those that did, including the scope boundaries and the questions that remain open.

2. Hypotheses and What Would Falsify Them

The three claims of Section 1 are operationalised as follows, and the criteria were fixed before the corresponding experiments were run.

- **H1, detection at parity.** Scored by F_1 at the 90th percentile of each series against labelled crisis windows, so that every metric spends the same alarm budget, with block-bootstrap intervals on the difference. Falsified if any baseline's advantage has an interval excluding zero. Three independent crisis definitions are used and reported separately, because a result that holds only under one labelling is not a result.

- **H2, attribution without a parameter.** Scored by the hit rate in the top- $|\text{epicentres}|$ on a synthetic benchmark where ground truth exists, across regimes with one to four simultaneous epicentres. Falsified if a spectral rule with a well-chosen k , or a published per-node alternative, scores higher across regimes.
- **H3, coincident register.** Scored by the cross-correlation between the index and an external stress index across lags. Falsified if the peak sits at a negative lag, which would make it a leading indicator.

A fourth question is not a hypothesis but a scope condition, and it is answered in Section 6.9: on what kind of network does the raw index work, and where does it need the persistence filter?

3. The Index

3.1. Construction on a Financial Network

For a rolling window of W days of daily log-returns of n assets, we build the correlation matrix ρ , define the adjacency matrix $A = |\rho|$ (absolute value, zero diagonal), and compute:

$$\text{TSI} = \frac{C \cdot D}{M} \cdot \text{Coex}, \quad (1)$$

where C is the weighted clustering coefficient (normalised $\text{Tr}(A^3)$), D is connection density, M is the inverse spectral gap of the graph Laplacian (a modularity proxy), and Coex is the variance of the degree sequence. The construction descends from prior work on co-inclusion and co-exclusion networks in agricultural microbiomes, where resilience indices were derived from network transitivity [4], following the original definition in [1]. Because C , D , M , and Coex all scale with n in different ways, raw TSI magnitudes are **not comparable across networks of different size**; only within-series dynamics (z-scores, percentiles) are.

The name. We call the quantity in (1) the **Triadic Stress Index** (TSI). Each part of the name records something that the rest of this paper either establishes or bounds. *Triadic*, because the closed-triangle count is the only one of the four factors that is not a function of the degree sequence alone, and because the per-node attribution of Section 6.1 is the triangle count read per node rather than summed. *Stress*, because in this domain the index rises under stress, which is the reverse of the orientation the same construction carries in the ecological setting it comes from, for the reason set out immediately below. And *index* rather than signal or forecast, because Section 6.7 establishes that it is coincident with stress and does not lead it.

A note on notation, because the names in this family are easy to confuse. The accompanying theoretical work writes this quantity Ω , and the machine-learning line built on the same trace carries the Omega name as well. To keep the financial instrument distinct from both, we write TSI throughout this paper, in prose and in equations, and reserve the other names for what they denote: **FSRI** for the theoretical framework, **Omega-S** for the machine-learning line, and **TSI** for the index defined here.

Relation to the composition fixed in the framework paper. Equation (1) places Coex in the numerator, whereas Definition 1 of the framework paper [1] places it in the denominator, so that a high degree variance lowers the index there and raises it here. The two agree on the other three factors, including the orientation of M as an inverse spectral gap. We keep the composition above, and we give the reason rather than leaving the difference to be found by comparing code.

The reason is that the two domains do not agree on what the same structure means. Under stress, correlations rise, the network becomes denser and more connected, and every factor moves in the same direction: C rises, D rises, Coex rises, and M , being an inverse spectral gap, falls. A stress indicator should therefore place the three that rise in the numerator and the one that falls in the denominator, which is what (1) does; the composition of Definition 1 places Coex against the other three. The underlying asymmetry is that a densely interconnected, weakly modular community is the healthy configuration in the ecological setting the index derives from, and is the crisis configuration in a market, where it means that everything falls together. The difference between the two papers is thus a deliberate reorientation by domain and not an unreconciled discrepancy.

We also tested it rather than only arguing it, on the same windows and with the same detection protocol used throughout this paper. Over a portfolio of large US banks from 2006 to 2011, with the crisis window labelled as 2007-08 to 2009-06, the filtered F_1 at the ninetyeth percentile is 0.275 for the composition of (1) and 0.157 for that of Definition 1, against a chance level of 0.152 for an alarm rate of ten percent at the observed crisis frequency. The composition used here detects at roughly $1.8\times$ chance while the alternative sits at chance. Two limits should be read alongside that. Block-bootstrap confidence intervals on the paired difference are wide and include zero ($+0.054$, $[-0.229, +0.315]$), so the comparison is directional rather than conclusive; and on this particular portfolio and period the Absorption Ratio also sits near chance (0.167), which indicates that the test itself discriminates weakly rather than that the alternative composition is uniquely poor. The argument from domain orientation above is the stronger of the two and does not depend on this measurement.

A third composition deserves reporting because its outcome is a limitation of the index rather than a point in its favour. The framework paper shows that $\text{Coex} = \text{Var}(k)$ carries units of degree squared and proposes a shape-normalised variant that replaces it by the Gini coefficient of the degree sequence and the density by the mean degree, removing the dependence on the scale of the degree sequence. On the same test that variant scores 0.098, that is *below* chance. The implication is uncomfortable and we state it: a substantial part of what TSI detects in this setting is the level of correlation rather than the shape of its distribution, and removing the scale dependence removes the signal with it. This does not affect the comparisons reported below, which are all against external baselines computed on the same windows, but it does bound what the index should be claimed to measure.

The closest structural precedent, and we do benchmark against it. Sandhu, Georgiou and Tannenbaum [5] compute Ollivier-Ricci curvature on correlation networks of equities and report it as an indicator of market fragility and systemic risk. That is the same graph construction and the same purpose as the present work, reached from discrete geometry rather than from triadic structure, and it is the nearest precedent to what follows. Earlier versions of this paper named it and declined to benchmark against it, on the grounds that per-edge Ollivier-Ricci requires a Wasserstein computation for every edge and therefore sits in a different cost regime from a stochastic trace estimate. That was a reason not to have done the work, not a reason for it not to be done, so we have now done it: Section 6.5 reports the head-to-head on the same windows, under the same filter and the same alarm budget as every other benchmark here. We note in passing that a subsequent analysis [6] argues the curvature identifies the size and duration of a crisis accurately without leading it, which is the same register we establish for TSI in Section 6.7.

Other network-structural indicators, and where this sits. Systemic risk has been approached through exposure and connectedness models [18, 19, 10] and through random-matrix cleaning of correlation matrices [16, 17]. The present work belongs to neither: it fixes a single composite index in advance, taken from an external domain, and asks how it scores against

spectral baselines on the same windows. Three lines of work are close enough that we state the relation explicitly rather than let a reader find it. Samal, Kumar, Yadav and Chakraborti [20] evaluate a battery of network-centric indicators, including edge curvatures, clique number and community counts, on threshold networks over minimum spanning trees of 69 global indices, and conclude that network measures are useful for monitoring fragility. That is the same question as ours; what differs is that they survey measures and we commit to one in advance. Bartesaghi, Diaz-Diaz, Grassi and Uberti [21] use the global balance index of the *signed* correlation network as a systemic-risk measure and derive a local, per-node version of it, developed further into an investment-decision rule in follow-up work [22]. That is the closest existing work to the signed extension this paper leaves open, and we state the position plainly: our $A = |\rho|$ discards precisely the information their index is built from, their global balance is already benchmarked against spectral risk measures as ours is, and their local balance is a per-node attribution of the same kind as ours. Rather than list the signed variant as future work, we ran it, and report three findings in Section 6.13. Zhang, Wang, Zheng and Cartledge [23] pair a correlation fragility indicator computed on time-varying correlation networks with a risk contribution score that attributes systemic risk to categories of protocol; a structural index plus a per-node decomposition is the same shape as what is reported here, on a different asset universe. None of the three is used as a running benchmark, for the cost-regime reason given above, and we regard a comparison against the signed balance index as the most informative of the three left undone.

4. Benchmarks

Four families of comparison are used, and the reason for each differs.

- **The Absorption Ratio**, described below. It is the industry standard and therefore the measure a practitioner would already have running. It is a weak baseline on this data, which the results below make plain.
- **The effective rank and the Vendi score**, spectral entropies of the correlation eigenvalues. These are the *sharpest* baselines we could find rather than the most common ones, and they are the honest test of H1. They are described and scored in Section 6.2.
- **Per-node alternatives**: spectral attribution under four different rules for choosing the number of components, the node degree, and the local balance index of Bartesaghi et al. All are scored in Section 6.1.
- **The global balance index** of the signed correlation network, treated separately below because it cannot be scored fairly on the networks used here.

4.1. Benchmark: the Absorption Ratio

The Absorption Ratio [7] is the fraction of total variance explained by the first k principal components of the return covariance matrix ($k = \lfloor n/5 \rfloor$). It is the de facto standard in systemic-risk management, used by MSCI and central banks, and is computed on the same rolling windows as TSI for direct comparability.

Additional spectral benchmarks. The Absorption Ratio summarises the eigenvalue spectrum through the mass of its leading components. Because TSI is itself partly a spectral object, benchmarking against the AR alone risks comparing against a deliberately coarse summary. We therefore add two sharper spectral baselines on the same windows: the *effective rank* [13], $\exp(H)$ with H the Shannon entropy of the correlation eigenvalues normalised to sum to one; and the *Vendi score* [14], the exponential of the von Neumann entropy of the normalised correlation kernel, which on a correlation matrix is *mathematically identical* to the effective rank, an equivalence we state explicitly rather than presenting the two as independent evidence.

We additionally report an effective rank computed on the return matrix itself (entropy of the singular-value spectrum), a genuinely distinct quantity.

4.2. The Global Balance Index, and Why It Is Not in the Main Table

The nearest published rival is the global balance index of the signed correlation network [21], $\kappa = \text{tr}(e^A)/\text{tr}(e^{|A|})$, with a per-node counterpart used for investment decisions in follow-up work [22]. It belongs in this list on merit: it is built on the same object, aims at the same target, and is published in a peer-reviewed venue. We reimplemented it and verified our version against the authors’ own public code, obtaining agreement to machine precision over 2232 windows.

It does not appear as a column in the detection table for a reason we measured rather than assumed. On small networks the index *saturates*: with eight nodes it sits at its maximum of 1 on 26% of windows in its weighted form and 70% in the binary form the same authors report, because a small correlation network is close to balanced whatever the market is doing. Any ranking produced under those conditions describes a degenerate regime, not the index, so we produce none. Section 6.13 reports what we did measure instead, including the regime in which the index is well behaved, a check that its own published relationship reproduces on our data, and a head-to-head on a 464-asset universe where it does not saturate.

5. Data and Protocol

5.1. Data

Table 1: Datasets used in this study.

Scenario	Assets (n)	Period	Source
Banking crises (Bear Stearns, Lehman, European debt crisis)	10 banks	2006–2011	yfinance
AI sector	11 stocks	2021–2026	yfinance
Out-of-sample validation	8 OFR components	2000–2026	OFR
Large-network robustness	42 stocks, 7 sectors	2006–2026	yfinance
Cross-market (crypto, FX, commodities, sovereign debt)	10+7+10+10	2021–2024	yfinance

All windows: 20 trading days, recomputed every 3 days.

5.2. TSI with Persistence Memory

We apply an asymmetric exponential filter:

$$\text{mem}(t) = \alpha_{\uparrow} \text{TSI}(t) + (1 - \alpha_{\uparrow}) \text{mem}(t - 1) \quad \text{if } \text{TSI}(t) > \text{mem}(t - 1), \quad (2)$$

$$\text{mem}(t) = \alpha_{\downarrow} \text{TSI}(t) + (1 - \alpha_{\downarrow}) \text{mem}(t - 1) \quad \text{if } \text{TSI}(t) \leq \text{mem}(t - 1), \quad (3)$$

with $\alpha_{\uparrow} = 0.6$ and $\alpha_{\downarrow} = 0.08$, chosen by inspection and later validated by grid search (Section 5.3).

5.3. Out-of-Sample Calibration and Significance Testing

Hyperparameters were calibrated via grid search ($\alpha_{\uparrow} \in [0.3, 0.8]$, $\alpha_{\downarrow} \in [0.02, 0.20]$) with a genuine temporal train/test split on the 2000–2026 OFR Financial Stress Index series (train 2000–2015,

test 2016–2026). Statistical significance of the F1 gap between TSI with memory and the Absorption Ratio was assessed via block bootstrap (block length ≈ 1 quarter, $B = 2,000$ resamples), restricted to the out-of-sample period only.

5.4. Persistent Homology

We implemented Vietoris-Rips persistent homology from scratch (boundary-matrix reduction, $\mathbb{Z}_2$ coefficients), validated against an analytical test case (a 6-point circle, which yields exactly one H1 generator). We tested two constructions:

Correlation-network H1 (negative result): Vietoris-Rips filtration on the asset correlation-distance matrix $d_{ij} = \sqrt{2(1 - \rho_{ij})}$ [9].

Takens-embedding H1 (positive result): following Gidea & Katz [8], a scalar stress series is reconstructed as a point cloud in $\mathbb{R}^d$ via time-delay embedding, and persistent homology is computed on that point cloud within each rolling window.

6. Results

What follows is ordered by argument rather than by chronology: first the layer that has no equivalent among the baselines, then the layer where the index merely matches them, then the comparison it wins and the one it does not, then the boundary of what it can be asked to do, and finally the robustness checks and the extension we closed with a null.

6.1. On the Node-Attribution Layer

Detection says the system is stressed. Attribution says *where*. The two are not equally hard, and the difference is not that spectral methods cannot attribute, because they can: the loading on the leading eigenvector, directly available to any Absorption-Ratio user, recovers a single epicentre perfectly. The difference is that they need to be told how many places to look, or given a rule for deciding it. Reading eigenvectors requires committing to k , the number of components to inspect, and in deployment k is either fixed in advance, before anyone knows how many epicentres the next episode will have, or selected per window by a rule that can itself be wrong. The per-node decomposition $\text{diag}(A^3)$, which is the alarm itself read per node rather than summed, has no such parameter. How much that is worth is the question this section answers, and the answer is more modest than the framing above suggests.

We tested this on synthetic networks with a known epicentre, in two steps. First, whether the attribution works at all: over 80 simulations the culprit rule recovers the true epicentre in essentially every run (top-3 hit rate 1.00 against a chance level of 0.25), and under two simultaneous uncorrelated epicentres it recovers 0.93 of the six affected nodes (chance 0.50). Then the question that decides the comparison, which is robustness when the number of epicentres is unknown. We ran regimes of one to four simultaneous epicentres, holding the spectral method at a fixed $k = 2$ as it would be held in deployment (120 simulations per regime, paired Wilcoxon):

Epicentres	Triangles		Spectral, by how k is chosen			
	$\text{diag}(A^3)$	Degree	fixed $k = 2$	MP	Kaiser	oracle
1	0.994	0.933	0.756	0.997	0.000	0.989
2	0.978	0.940	0.999	0.978	0.071	0.999
3	0.980	0.952	0.946	0.962	0.457	0.981
4	0.975	0.961	0.948	0.959	0.808	0.989

Top-|epicentres| hit rate, 120 simulations per regime, 16 nodes and 40-observation windows. *MP* selects k per window as the number of eigenvalues above the Marchenko-Pastur edge [15]

$(1 + \sqrt{n/T})^2 = 2.665$, the standard random-matrix rule; it picks $k = 1.71$ on average. *Kaiser* selects the eigenvalues above 1 and picks $k = 5.87$. *Oracle* is given the true number of epicentres and is not available in practice; it is the ceiling.

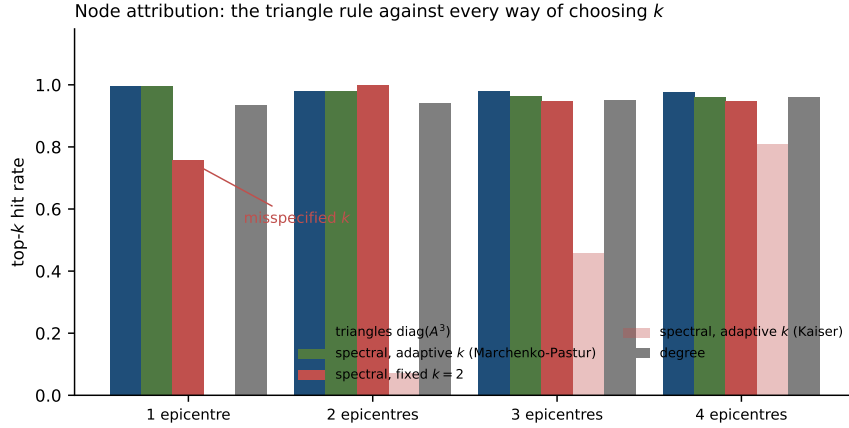


Figure 1: Attribution accuracy as the number of simultaneous epicentres varies. The triangle rule is flat across regimes because it has no parameter to misspecify. The spectral rule depends entirely on how k is chosen: the Marchenko-Pastur rule ties the triangle rule everywhere, a fixed $k = 2$ costs a quarter of the accuracy when it is wrong, and the equally standard Kaiser rule collapses. The spread between those three, not the gap to the triangle rule, is the result.

The triangle rule is flat, 0.975 to 0.994 across every regime, because there is no parameter in it to misspecify. Fixed $k = 2$ matches it exactly where that value happens to be right (two epicentres) and loses a quarter of its accuracy where it is not (0.756 with a single epicentre).

The adaptive- k control, and what it costs the claim. An earlier version of this section stopped there. It should not have, because the obvious objection is that a practitioner would not hold k fixed but read it off the spectrum window by window, and we had not tested it. We have now, and the result is against us: with k chosen by the Marchenko-Pastur edge the spectral rule scores 0.959 to 0.997 and ties the triangle rule across all four regimes. The oracle ties it as well. The gap in the fixed- k column is therefore not evidence that triangles see something eigenvectors cannot; it is evidence that k matters, and a good rule for choosing k closes it.

What the control does not do is make the choice free. The three selection rules in the table span 0.000 to 0.997 on the single-epicentre regime: Marchenko-Pastur lands near the ceiling, a fixed $k = 2$ costs a quarter of the accuracy, and the Kaiser rule ($\lambda > 1$), which is equally standard and equally defensible a priori, collapses completely, because it selects roughly six eigenvectors and the epicentre stops being separable in their combined loading. The spectral route can match $\text{diag}(A^3)$, but only conditional on a selection decision whose wrong answers are catastrophic and are not identifiable without the ground truth we have here and a practitioner does not.

So we state the claim at the size the evidence supports, which is smaller than the one we set out to make. TSI’s attribution is not more accurate than spectral attribution; against a well-chosen adaptive rule it is exactly as accurate. What it is, is unconditional: it arrives with the index, ranks the nodes with no parameter to select and no way to select it wrongly, and it also beats the simpler degree rule in all four regimes (0.933–0.961), which is why we adopt it over a generic centrality. That is a robustness and simplicity argument, not a resolution argument, and it should be read as one.

Two further limits bound even that. The evidence is synthetic, with a controlled ground truth real markets do not provide, so it is a mechanism result and not a field validation. And it does not overturn the detection tie of Section 6.2: on discriminating stressed windows from calm ones, TSI and the effective rank remain statistically indistinguishable, and nothing here

changes that.

6.2. Benchmarking Against Sharper Spectral Baselines

Because the Absorption Ratio is a deliberately coarse spectral summary, we re-ran the identical protocol (same windows, crisis labels, F1-at-90th-percentile criterion and block bootstrap) against the sharper baselines of Section 4.1. One design point decides this comparison and we handle it explicitly: the persistence filter of Section 5.2 helps any series that is elevated in bursts, so filtering TSI and not its rivals would flatter TSI. We therefore report every metric twice, raw and under the same asymmetric filter, and run the bootstrap on the like-for-like column. Out of sample (2016–2026, 897 windows):

Metric	F1@p90		$\Delta F1$	95% CI	p	Verdict
	raw	filtered				
TSI	0.367	0.447	—	—	—	—
Effective rank (corr.)	0.347	0.377	+0.038	$[-0.010, +0.111]$	0.10	tie
Vendi score	0.347	0.377	+0.037	$[-0.009, +0.104]$	0.12	tie
Absorption Ratio	0.174	0.134	+0.219	$[+0.070, +0.410]$	< 0.0005	TSI wins
Effective rank (returns)	0.179	0.174	+0.227	$[+0.097, +0.393]$	< 0.0005	TSI wins

$\Delta F1$ and its interval come from the block bootstrap on the filtered column and need not equal the difference of the two point estimates. The filter is the same asymmetric map for every row, applied to the stress-oriented signal (the effective rank and the Vendi score fall under stress, so they enter negated).

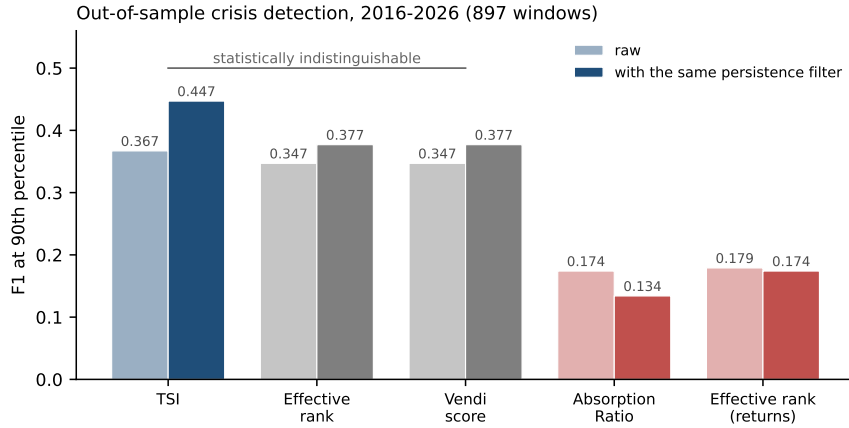


Figure 2: Out-of-sample crisis detection, each metric read raw (pale) and under the same persistence filter (solid). TSI outperforms the Absorption Ratio by a wide margin in both readings, but the sharper spectral baselines sit within its confidence interval in both. The advantage over the industry standard is real; the advantage over a properly sharpened spectral measure is not, and giving the rivals the same filter does not change that verdict.

The verdict does not depend on which column is read. Raw, TSI leads the effective rank by 0.020; filtered, by 0.070; in neither case does the bootstrap interval exclude zero. Over the full history (2000–2026, 2,232 windows) the pattern repeats: under the same filter TSI scores 0.389 against 0.342 for the effective rank (tie, $p = 0.07$) and 0.175 for the Absorption Ratio (TSI wins, $p < 0.0005$).

Does the verdict depend on the crisis list? For one comparison, yes. Re-running the same protocol under each of the three lists of Section 6.3, on both the hold-out period and the full history, with every series filtered:

Sample	List	TSI	Eff. rank	Absorption R.	Eff. rank (ret.)
Out of sample 2016–2026	16 ep. (21.9%)	0.434	0.343 =	0.168 =	0.224 •
	23 win. (34.9%)	0.447	0.377 =	0.134 •	0.174 •
	25 ep. (33.0%)	0.451	0.383 =	0.135 •	0.181 •
Full history 2000–2026	16 ep. (27.7%)	0.340	0.363 =	0.204 =	0.192 =
	23 win. (40.6%)	0.389	0.342 =	0.175 •	0.165 •
	25 ep. (39.8%)	0.386	0.343 =	0.164 •	0.167 •

= tie, • TSI wins, by block bootstrap on the F1 gap. Percentages are the share of windows the list labels as stress.

Two readings follow. The tie against the effective rank is *robust*: it holds in all six cells. Its direction is consistent, with TSI ahead in five of the six, and in the sixth the effective rank leads by 0.023, well inside the noise ($p = 0.59$). The win over the Absorption Ratio is *not* robust: it disappears in both cells of the narrowest list, where the gap remains positive (+0.170 and +0.121) but the interval touches zero. We report this because the Absorption Ratio is the benchmark this work set out to beat and the one a practitioner would already have in production. The correct statement is that TSI beats it clearly under the two broader labellings and falls short of conventional significance under the narrowest, which is also the one with the fewest labelled windows and the least power.

Why the tie happens, and what it is not. The four baselines are not four independent objects. The Vendi score and the effective rank are identical by construction on a correlation matrix, as noted in Section 4.1, and empirically the Absorption Ratio and the effective rank computed on the return matrix are near-duplicates as well (Pearson 0.91, Spearman 0.97), which is why they score alike. That leaves two distinct baselines, not four. Against them TSI is far from equidistant: it correlates 0.85 (Spearman 0.93) with the effective rank and only 0.54 (Spearman 0.61) with the Absorption Ratio, and it shares 53% of its alarm windows with the former against 34% with the latter. TSI is therefore a close relative of the effective rank and a distant one of the Absorption Ratio, which is exactly what the companion theoretical work predicts: the global scalar $\text{Tr}(A^3)$ is the third spectral moment and belongs to the same family as the spectral entropies [1], whereas the Absorption Ratio reads leading-eigenvalue mass. The tie is not an awkward null result to be explained away; it is the predicted outcome, and the 47% of alarms TSI does not share with its nearest relative is the measure of how far a different construction of the same spectrum can move.

Alarm quality on a common criterion. Section 6.3 gave an unexplained-alarm rate for TSI alone. Computed for every metric on the same 25-episode list, at the same top-decile alarm budget (full history, all series filtered): TSI 4.0% (9 of 224), effective rank and Vendi 14.7% (33), effective rank on returns 58.5% (131), Absorption Ratio 59.4% (133). Out of sample the ordering is the same (3.3%, 17.8%, 61.1%, 71.1%). TSI’s alarms are therefore the cleanest of any metric tested, by a wide margin over the industry standard and by roughly a factor of four over the effective rank.

One qualification prevents that from being read as an independent result, and it is the reason we ran the comparison rather than quoting the 4% on its own. With the alarm budget fixed at the top decile, the unexplained-alarm rate is exactly $1 - \text{precision}$, and F1 at a fixed alarm count is a monotone function of the same quantity. The two comparisons are one comparison in two presentations. The verdict from the bootstrap therefore carries over unchanged: against the effective rank the difference in alarm quality is real in point estimate and not separable from zero at this sample size, exactly as the F1 comparison found.

We read this as the honest and more informative result. TSI’s advantage over the Absorption Ratio is real, reproducible and statistically significant but the gap is largely explained by the AR being a weak baseline on this data ($F1 \approx 0.17\text{--}0.19$, roughly half of every other measure tested), not by TSI accessing information the spectrum lacks. **Against a properly sharpened**

spectral baseline, TSI ties. It does not lose, and retains a small positive point estimate in every comparison, but the difference is not statistically distinguishable from zero. The defensible claim is therefore not that TSI dominates spectral methods, but that it matches the best of them while being constructed from a different primitive, network topology rather than variance decomposition, and while carrying an attribution layer that arrives with the index and needs no selection rule (Section 6.1, which also reports what that is and is not worth). A lead-lag analysis (Section 6.7) establishes the correct register for all of these claims: the cross-correlation with the stress index peaks at zero lag and decays symmetrically, so TSI is a *coincident state index*, reporting that the network is presently in a concentrated and stressed configuration, and not a leading indicator. We frame every claim accordingly.

6.3. Out-of-Sample Calibration and False-Alarm Rate

Over the full 2000–2026 OFR series (2,232 windows), the correlation between in-sample and out-of-sample F1 across the full 36-combination grid is **0.96**. The optimum ($\alpha_{\downarrow} = 0.08$, robust for $\alpha_{\uparrow} \in [0.5, 0.7]$) matches the values chosen by inspection.

Three ground-truth lists, and which figure uses which. The episodes that count as stress were labelled at three points in this project and the lists are not identical. We set them out rather than let a reader assume a single list throughout, because each headline number below is attached to a different one.

- A *16-episode* list with narrow windows, used for the hyperparameter calibration of Section 5.3. On it, out of sample (2016–2026), TSI with memory gives **69% precision and 32% recall** at the 90th-percentile threshold.
- A *23-window* list with wider windows, used for every F1 figure in Sections 6.4 and 6.2. On it, out of sample, the same series gives precision 1.00 and recall 0.29 (90 alarms, none falling outside a labelled window).
- A *25-episode* list used to ask a different question: not whether an alarm falls inside a pre-drawn window, but whether it corresponds to any documented episode at all. Across the full history 9 of 224 alarms have no such correspondence, a **4.0% unexplained-alarm rate**.

Three caveats bound that last figure and we state all three. It is computed over the full 2000–2026 history rather than the hold-out period, so it is not an out-of-sample result. Its list is broader than the 23-window one, so it is a laxer test by construction. And we have not computed the same quantity for the Absorption Ratio or for the spectral baselines, so it characterises TSI and does not separate it from them. Rather than pick one list and discard the others, we re-ran every comparison under all three. Section 6.2 reports the result: one verdict in this paper does depend on which list is used, and it is not the one we expected.

6.4. Statistical Significance

Restricted to 2016–2026 (897 windows), the F1 gap between TSI with memory and the Absorption Ratio is 0.273 (0.447 vs. 0.174). Block-bootstrap ($B = 2,000$, block ≈ 1 quarter): **95% CI** [0.095, 0.392], does not cross zero, one-sided $p < 0.0005$.

TSI with memory here is the asymmetric filter of Section 5.2 ($\alpha_{\uparrow} = 0.6$, $\alpha_{\downarrow} = 0.08$) applied to raw TSI, which is the form used throughout this paper. We say so explicitly because the next section compares against spectral rivals and the choice of filter moves the number materially.

Caveat. This metric rewards sustained-plateau behaviour, which is precisely what the memory filter was built to produce. The test confirms the filter achieves its design goal with statistical confidence, not that TSI with memory is superior on some independent criterion.

The head-to-head, run where the index is well behaved. On 464 S&P 500 constituents with a complete ten-year history the balance index does not saturate: it takes distinct values in every window, and at the 90th percentile it flags 10.2% of them, as do TSI, the binary balance index and the effective rank. The Absorption Ratio cannot be scored in this configuration at all, and for a reason worth stating: with a 60-day window and 464 assets the covariance matrix has rank 59, so the top $n/5 = 93$ eigenvalues capture the entire trace by construction and the ratio is identically one. That is a limitation of the configuration, not of the measure.

Ground truth here is *theirs*, not ours: a window is labelled systemic when the cross-sectional mean return falls below a threshold over a 20-day span, which is the definition used in the original work. Adopting it removes our own episode judgement from the comparison entirely.

The result is a tie, at every threshold and under both readings of the label. Scoring against the contemporaneous span, TSI reaches F_1 of 0.160–0.161 and the balance index 0.184–0.187, with every paired difference’s interval crossing zero (p from 0.43 to 0.63); the binary variant and the effective rank land in the same place. No measure separates itself, and none is far above the chance level of roughly 0.14 at this alarm budget and base rate. We read that as a property of the label rather than of the measures: a return-based event is a coarse proxy for a structural one, and this is the price of using someone else’s definition rather than our own.

One thing did fall out of it, and it corroborates the central claim of this paper from an unexpected direction. Orientation was measured rather than assumed in every cell. Against the *contemporaneous* span TSI enters positively, rising as returns deteriorate. Against the *forward* span its orientation reverses. An index that reads the present correctly and inverts when asked about the future is precisely a coincident state index, which is what Section 6.7 establishes by a different route. These tests use a different universe and a different ground truth from Section 6.2, so they form their own family and are not folded into the correction of Section 6.6.

6.5. Head-to-Head Against Ollivier–Ricci Curvature

Correlations are mapped to the Mantegna distance $d_{ij} = \sqrt{2(1 - \rho_{ij})}$, each node carries a measure over its neighbours proportional to the absolute correlation, and the curvature of an edge is $\kappa_{ij} = 1 - W_1(m_i, m_j)/d_{ij}$ with the earth-mover distance computed exactly by linear programming. The market-level scalar is the mean edge curvature; a correlation-weighted mean was also computed and behaves almost identically. Everything else is unchanged: same windows, same asymmetric filter applied to the curvature series as to every other, F_1 at the 90th percentile, block bootstrap, all three labellings.

Orientation was measured rather than assumed. Mean curvature is 0.711 in labelled crisis windows against 0.678 in calm ones, so it rises under stress, which is the direction the original work reports.

Sample	Labels	F_1 TSI	F_1 curvature	ΔF_1 (p)
OOS 2016–2026	16 episodes	0.434	0.301	+0.126 (0.011)
OOS 2016–2026	23 windows	0.447	0.337	+0.087 (0.007)
OOS 2016–2026	25 episodes	0.451	0.321	+0.101 (0.003)
FULL 2000–2026	16 episodes	0.340	0.335	+0.023 (0.388)
FULL 2000–2026	23 windows	0.389	0.317	+0.074 (0.018)
FULL 2000–2026	25 episodes	0.386	0.316	+0.073 (0.023)

Curvature lands where a reader familiar with this literature would expect it to land: clearly above the Absorption Ratio, which scores 0.13 to 0.17 on the same windows, and somewhat below the effective rank, which scores 0.34 to 0.38. It is a mid-strength baseline rather than a weak one, and the honest summary is that TSI is ahead of it in all six cells but not by the margin it holds over the Absorption Ratio. Section 6.6 reports what survives once this comparison is folded into the corrected family, and only one of the five nominally significant cells does.

6.6. Correcting for Multiple Comparisons

This paper runs many related tests, and reporting each in isolation would overstate the evidence. We therefore define the family in advance and correct over all of it. Hypothesis H1 is tested by comparing TSI against every *distinct* baseline (the effective rank, which is identical to the Vendi score on a correlation matrix and so is counted once; the Absorption Ratio; and the effective rank computed on the return matrix), on both samples, under all three crisis labellings. With the Ollivier–Ricci curvature of Section 6.5 included, that is $4 \times 2 \times 3 = 24$ tests, and all twenty-four enter the family, including those that were never going to be reported as wins: choosing which tests belong is precisely the manoeuvre the correction exists to prevent. We apply Holm–Bonferroni at a family-wise error rate of 0.05, which is uniformly more powerful than plain Bonferroni and assumes no independence between tests.

Sixteen of the twenty-four are significant before the correction and nine after it.

The nine that survive are the eight wins over the Absorption Ratio and over the effective rank on returns, under the 23-window and 25-episode labellings, on both samples, all with bootstrap p below 0.0001 against Holm thresholds between 0.0028 and 0.0045; plus one of the six comparisons against curvature. The six comparisons against the effective rank were ties before the correction and remain ties after it, so the central finding of this paper is untouched: a correction cannot turn a tie into anything else.

Three results are lost, and all three sit on the same labelling. The win over the effective rank on returns out of sample under the 16-episode list ($\Delta F_1 = +0.179$, $p = 0.0060$ against a threshold of 0.0050), and the wins over the Absorption Ratio and the effective rank on returns over the full history under the same list ($p = 0.040$ and $p = 0.044$). This is the narrowest of the three labellings and the one with the least power, and the paper already identified it as the weakest support for the Absorption Ratio result before the correction was applied. The correction did not reveal a new weakness; it quantified one that was already declared.

The practical reading is that the advantage over the Absorption Ratio is real under the two broader labellings and does not reach family-wise significance under the narrowest. We state the claim at that strength and no higher.

6.7. Lead–Lag Analysis: a Coincident Index, Not a Forecast

The F1-at-90th-percentile criterion used throughout asks whether TSI is *elevated during* documented stress windows. It is a contemporaneous classification test, and it says nothing about whether TSI rises *before* stress. Because the framing of a “fragility index” invites the stronger reading, we test the stronger claim directly.

We cross-correlate TSI against the OFR Financial Stress Index [12] at lags $k \in [-15, +30]$ observations (each step ≈ 3 business days), where $k > 0$ means TSI leads. Results on the full history ($n = 2,232$):

Signal	Peak $ \rho $ at	ρ at $k = 0$	Reading
TSI raw	$k = 0$	+0.203	exactly coincident
TSI with memory	$k = -2$	+0.195	coincident (marginally lagging)
Absorption Ratio	$k = -15$	+0.063	lagging

Both series are strongly autocorrelated, so the absolute ρ values in levels are not the object of interest; the informative feature is the *shape* of the cross-correlation function, which decays symmetrically about $k \approx 0$, the signature of a contemporaneous relationship rather than a leading one. Repeating the analysis on first differences sharpens the same conclusion: Δ TSI raw peaks exactly at $k = 0$ ($\rho = +0.094$), and at $k = +1$, with TSI leading by a single step, the correlation is statistically indistinguishable from zero ($\rho = -0.009$ and $+0.008$ for the memory and raw forms respectively, against a significance threshold of ± 0.041 at $n = 2,231$).

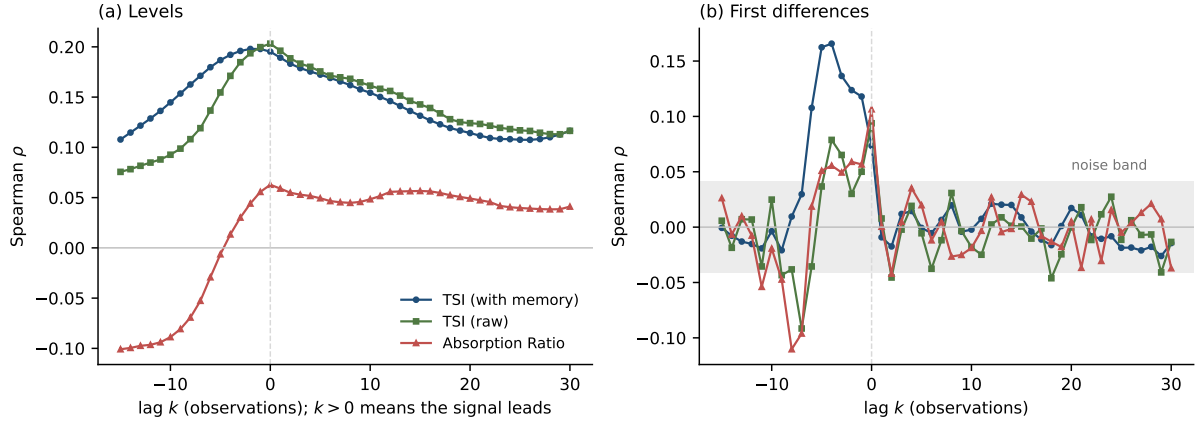


Figure 3: Cross-correlation against the OFR Financial Stress Index. TSI with its persistence filter is in dark blue, raw TSI in green, and the Absorption Ratio in red. **(a)** In levels, TSI peaks at or just before zero lag and decays symmetrically. **(b)** In first differences the same conclusion sharpens: at positive lags, where a leading indicator would show its peak, the correlation lies inside the noise band. The shape identifies a coincident state index rather than a forecast.

The memory filter, being an exponential moving average, introduces a small lag rather than an anticipation, peaking at $k = -4$.

What this does and does not change. It does not affect the paper’s principal result: the F1 comparison was always a contemporaneous test, it reproduces, and TSI outperforms the Absorption Ratio at every lag examined ($\rho \approx 0.20$ versus 0.06). What it does change is the permissible framing. TSI should be read as a *state index*, closer in spirit to Convective Available Potential Energy in meteorology, which reports that the atmosphere is presently primed for severe weather without predicting when or where a storm will form, than as an early-warning signal. A high reading says the correlation network is presently in a concentrated, stressed configuration. It does not say that stress is coming. We adopt that language throughout and recommend that practitioners do the same.

6.8. Four Original Scenarios

Bear Stearns (March 2008). Raw TSI peaks *before* the rescue and falls afterward, while the Absorption Ratio keeps rising; TSI with memory tracks the AR’s sustained trajectory.

Lehman Brothers (September 2008). Raw TSI jumps sharply within days of the collapse, a much sharper reaction than the AR, and TSI with memory preserves this signal through October.

European debt crisis (2011): the original failure case. Raw TSI falls to 0.17–0.30 during the crisis on the 10-bank network while the AR stays elevated; TSI with memory corrects this. Section 6.9 shows this failure does not replicate on a larger, diversified network.

AI sector: the 2025 tariff shock. The same failure-and-correction pattern recurs on the tech-stock network ($n = 11$).

6.9. Large-Network Robustness and the Central Finding

Running the full pipeline on a 42-asset network diversified across seven sectors, 2006–2026, and normalising to z-scores:

The pattern sits in the epicentre column. During the European debt crisis the ten-bank basket, which *is* the epicentre sector, ends at $z = -1.09$, below its own historical mean while the crisis is still running, whereas the diverse and full baskets end at $+5.35$ and $+2.13$; the

Table 2: Ending z-scores by network composition.

	Financials ($n = 10$, epicentre)	Tech ($n = 8$, non-ep.)	Diverse ($n = 9$, mixed)	Full ($n = 42$)
Bear Stearns	−0.28	−0.42	−0.26	−0.12
Lehman	0.10	1.20	1.27	2.61
Euro debt	−1.09	2.18	5.35	2.13

same ordering holds at Lehman. That collapse of the reading during the episode is the forgetting behaviour the memory filter was built to repair, and it appears only where the basket is concentrated on the epicentre.

Node count alone does not explain it: the 8-node tech-only basket does **not** exhibit the forgetting pattern. **The variable that matters is whether the tested basket is concentrated in the sector that is the actual epicentre of the crisis.** Any basket that avoids that concentration, whether small-and-diverse or large-and-diverse, retains an elevated reading.

Replication in COVID ($z = 1.35$, Feb–Apr 2020) and the 2022 rate-hike selloff ($z = 1.41$, Jan–Oct 2022) confirms the pattern in two independent episodes.

6.10. Cross-Market Test: Crypto, Commodities, FX, and Sovereign Debt

Two clear successes. The all-time peak in the crypto series ($z = 3.9$) falls in May–June 2022, the Terra/Luna collapse, with MATIC-USD flagged as the most-connected asset. The all-time peak in sovereign debt ($z = 3.6$, August–December 2023) coincides with the historic bond selloff, with MBB (the highest-duration instrument) flagged as responsible.

Two episodes outside what TSI is built to measure. The September 2022 pound sterling / UK gilt crisis: TSI with memory stayed negative throughout, because the crisis was largely contained to sterling-linked instruments rather than reorganising the broader FX correlation structure. The 2022 Ukraine invasion: a ~ 3 -month lag before TSI rose, consistent with geopolitical shocks propagating through price co-movement over weeks rather than instantaneously.

These are not detection errors but *scope boundaries*: TSI detects correlation-structure breaks (episodes where normally semi-independent assets suddenly co-move), not directional price events. This yields a precise, falsifiable scope claim.

6.11. Attempted Replication on Shorter Series

We attempted to repeat the calibration on the bank network (train: Bear Stearns + Lehman 2008; test: European debt crisis 2011) and the AI network. Both failed to reproduce the OFR result, and not because the filter stopped working, but because the highest TSI with memory readings in the training period systematically fell *outside* our defined crisis windows (they clustered on real but undocumented events).

Finding. With short series (5 years, 2–3 labelled episodes), the definition of the ground-truth event list dominates the calibration result far more than the choice of hyperparameters. The 26-year, 20+-episode OFR series has enough independent events to support reliable out-of-sample calibration; 5-year, 2–3-episode series do not.

6.12. Persistent Homology

The correlation-network H1 construction showed no discriminative pattern (mean 0.03–0.07 across all scenarios; calm and crisis periods indistinguishable). We attribute this to insufficient topological richness in an 8–11-node graph.

The Takens-embedding construction tells a different story. The all-time maximum (30 March 2020, COVID) sits ~ 12 standard deviations above the historical mean (0.104 ± 0.110). **Fifteen of the twenty highest readings fall inside documented crises**, including October–November 2011, exactly the episode the small-network raw TSI missed.

6.13. The signed channel, and a head-to-head against the balance index

Three questions follow from the previous paragraph, and we answer them on the same windows, with code in the repository.

First: does the sign of the triangle carry independent information? No. Replacing $\text{Tr}(A^3)$ by $\text{Tr}(\rho^3)$, which counts balanced minus frustrated triangles, changes nothing: the two series correlate at 1.000 across all 2,232 windows, and the resulting F_1 moves by 0.000 (95% CI $[-0.005, +0.005]$). The reason is structural rather than particular to this data. Positive semi-definiteness bounds how frustrated a triangle in a correlation matrix can be, so signed and unsigned triangle counts are nearly collinear by construction; the measured ratio averages 0.92. The sign channel that the balance literature exploits is not reachable through the triangle count alone, and we retire this extension rather than carry it forward.

Second: the balance index as an attribution rule. The local balance of Bartesaghi et al. shares with $\text{diag}(A^3)$ the property of having no k to select, so it belongs in the comparison above on merit. On the same benchmark it scores 0.329, 0.552, 0.724 and 0.841 for one to four epicentres, against 0.992, 0.996, 0.972 and 0.966 for the triangle rule, with chance at 0.188 to 0.750. Because the generator contains no sign structure, which is precisely what a balance index is built to read, we repeated it with half of each stressed block loading negatively on its factor, so that frustrated triangles are present by construction. The ordering does not move: 0.992, 0.988, 0.979, 0.968 against 0.350, 0.567, 0.719, 0.838.

Third: detection, where we decline to claim a win. On our windows the balance index scores poorly, but the reason disqualifies the comparison rather than settling it. The index *saturates*: with 20-day windows over eight nodes it sits at its maximum of 1 on 26% of windows in its weighted form and 70% in the binary form the same authors report, because short-window correlation matrices are almost entirely positive and an all-positive network is perfectly balanced. On the 42-asset network of Section 6.9 lengthening the window makes it worse rather than better, the fraction of windows at $\kappa = 1$ rising from 10% at 45 days to 24% at 100 and 63% at 400.

That last observation, however, is a statement about small networks and not about the index, and the distinction decides who the finding is about. Repeating the measurement on 447 S&P 500 constituents, and again on 464 over a ten-year panel, the saturation disappears: κ ranges over almost the whole unit interval, with a median of 0.44 on the first panel and 0.03 on the second at a 60-day window, and at most 2.0% of windows sit at the maximum at any window length from 45 to 500 days. The measurement is reproduced by `code/validation/balance_saturation_sp500.py`. The reason is combinatorial. The number of triangles grows as n^3 , so in a large network a modest fraction of negative correlations already produces enough frustration to move the index, whereas a network of a few dozen nodes is close to balanced whatever the market is doing. Saturation is therefore a property of *small* correlation networks, and the regime the original authors work in, 199 to 385 assets, is not one of them.

Two consequences follow, and neither favours us. Our own networks of eight and 42 nodes are exactly the regime where their index cannot be evaluated, so no ranking produced on them would be informative about it, and none is given here. And the level of κ is not portable: the median moves from 0.44 to 0.03 between two panels of the same market a few years apart, so a fixed threshold calibrated on one dataset need not fire at all on another. What we do report is a check in the other direction: on our data their headline association reproduces, with the Spearman correlation between the balance index and mean absolute correlation at 0.94, 0.89 and 0.71 for the three window lengths against the 0.877 they report, and with the conditional

mean of forward average returns decreasing monotonically across balance quintiles at 45-day windows, as they describe.

A methodological note that this exercise forced, and that applies to the like-for-like table of Section 6.2. F_1 at a fixed percentile is a fixed alarm budget only if the metric takes distinct values near the threshold. A saturating metric flags far more than the intended fraction of windows and scores an inflated F_1 through recall alone; the binary balance index flags 70% of windows at its 90th percentile and reaches $F_1 = 0.537$ with a precision of 0.425 against a base rate of 0.406, which is no better than chance. We verified that no benchmark reported in this paper has this property. Every series in that table, raw and filtered, takes distinct values throughout and flags exactly 10.0% of windows at its 90th percentile; the check is printed by `code/validation/samefilter.benchmarks.py` in the repository.

7. Discussion

What the index is worth, stated as a package. No single comparison in this paper is decisive on its own, and we do not present one as such. Taken together they describe something narrower and, we think, more useful than a win. On detection, TSI matches the best spectral measure available: a tie in all six labelling-by-sample cells, with the point estimate in TSI’s favour in five of them, which is a consistent direction that never separates from zero at this sample size. At the same alarm budget its alarms are the cleanest of any metric tested, 4.0% unexplained against 14.7% for the effective rank and roughly 59% for the Absorption Ratio, though that is the same comparison re-expressed and carries the same verdict. It reaches that level from a different primitive, network topology rather than variance decomposition, sharing only half its alarms with its nearest relative. And it arrives with a node-attribution layer that requires no selection rule, ties the best adaptive spectral rule, and cannot be misconfigured the way a fixed or badly chosen k can be.

None of those four is a headline on its own. The defensible claim is the conjunction: *equal detection performance, cleaner alarms in point estimate, a different construction, and attribution included at no extra parameter*. A practitioner already running an effective-rank monitor would not switch on the strength of the detection numbers alone, and we do not suggest they should. What they would gain is the attribution layer, which their current tool supplies only after they commit to a component count and get it right.

The central methodological lesson is that **whether a tested basket is concentrated in the sector at the epicentre of a given crisis, not raw network size, determines whether raw TSI needs a memory correction**. On a network concentrated in the epicentre sector (10 banks during banking crises), raw TSI behaves as a sharp but forgetful alarm. On a network that avoids that concentration, whether large-and-diverse (42 assets) or small-but-diverse (9 assets spanning seven sectors), raw TSI already exhibits sustained-elevation behaviour, replicated across three independent crises.

8. Limitations

On real matrices the attribution reduces to the degree. The per-node layer is defended in this paper on a synthetic benchmark where the ground truth is known, and there $\text{diag}(A^3)$ scores 0.966 to 0.996 against 0.933 to 0.961 for the far simpler node degree, $\sum_j |\rho_{ij}|$. That gap does not survive contact with real data. Across 245 windows of 464 S&P 500 constituents, the cross-sectional rank correlation between $\text{diag}(A^3)$ and the degree averages **0.996**, and a cubic in the degree explains $R^2 = 0.992$ of the triangle count. We then asked whether the remaining 0.8% carries anything of its own, since that residual is the only part of the quantity that is not a restatement of the degree. It does not: its rank correlation with forward five-day returns is +0.002 with a bootstrap interval spanning zero, and its apparent association with

forward volatility (+0.187) disappears once current volatility is partialled out (+0.016, interval $[-0.003, +0.035]$), while the degree itself retains a small but non-zero effect there. The practical reading is that on correlation networks of real assets, naming the culprit with $\text{diag}(A^3)$ and naming it with the degree give the same answer, and a practitioner should use the degree, which costs one matrix row-sum. What the triangle count buys is what Section 6.1 claims and no more: no parameter to misspecify, and a measured advantage in a regime where the ground truth can be checked.

- The large-network result (Section 6.9) is a single test on one 42-asset universe; it has not been repeated with different sector compositions or other crises.
- The attempted short-series recalibration is a negative methodological result; we did not re-derive “correct” crisis windows to avoid indefinitely re-tuning ground truth.
- The Takens-embedding H1 result uses one embedding dimension and delay, chosen once; a proper replication would test robustness to these choices.
- The statistical significance test evaluates the memory filter against the exact objective it was designed for; it is not independent evidence of general superiority.
- Three different ground-truth lists are in use across the paper (Section 6.3). Every F1 figure uses one of them consistently, but the precision, recall and unexplained-alarm figures do not come from the same list as the F1 figures, and no result should be quoted as if they did.
- The attribution result of Section 6.1 is established on synthetic networks. Against a spectral comparator with an adaptive k it is a tie, not an advantage; the advantage claimed is the absence of a selection rule, not accuracy.
- The win over the Absorption Ratio does not survive the narrowest of the three crisis lists (Section 6.2), and it is one of the three results that a Holm–Bonferroni correction over the family of eighteen tests removes (Section 6.6). The tie against the effective rank survives all three lists and is unaffected by the correction, as a tie must be.
- Nothing in this document constitutes investment advice.

9. Where This Stands

Several questions that a reader would reasonably raise have already been answered, and it is more useful to say so than to list them as future work.

Settled, and reported above. Whether the conclusions survive a correction for multiple comparisons: Section 6.6 defines the family of twenty-four tests and applies Holm–Bonferroni, and reports what does not survive it. Whether the index beats the nearest structural precedent, Ollivier–Ricci curvature: Section 6.5 runs that head-to-head, which earlier versions declared and did not perform. Whether declining to rank the global balance index on small networks was concealing a loss: it was not, and Section 6.13 reports the tie on a 464-asset universe where the index is well behaved, under the original authors’ own definition of a systemic event. Whether the attribution advantage survives a spectral rule that chooses its own number of components: it does not, and the claim was reduced accordingly to robustness rather than resolution. Whether the alarm-quality figure holds for the competitors too: computed for all of them. Whether the results depend on which list of crisis episodes is used: all three are reported. Whether signing the correlation recovers the single-asset blind spot: it does not, and the reason is structural. Whether the per-node layer sees something the node degree does not: on real matrices it does not, and Section 8 says so with the numbers.

Open, and worth doing. A systematic sweep of the Takens embedding parameters, which would not change any conclusion here. And an attribution test on real data, which needs a partial ground truth, for instance from regulatory post-mortems of documented episodes; we note that after the degree result of Section 8 the expected outcome of that test is fairly clear, which lowers its value.

Not attempted here, deliberately. Everything in this paper is a backtest of a state reading. Turning that into an allocation rule is a different exercise with a different evidential standard, and mixing the two would weaken both.

Issues, pull requests and replication attempts are welcome at <https://github.com/BiomeMakers/TSI-OmegaS>. For production licensing: acedo@biomemakers.com.

10. Conclusion

TSI’s apparent need for a memory correction turns out to be largely a property of small, sector-homogeneous, or crisis-epicentre-concentrated networks, not of TSI itself. On a large, diversified network, raw TSI already tracks sustained stress correctly. The memory filter remains a validated, statistically significant, out-of-sample-tested fix for the small or concentrated-network case, which is the realistic case for many practical applications (a single sector, a single portfolio, a single institution’s counterparties).

Across all tests in this paper, namely four original scenarios, out-of-sample calibration, statistical significance testing, the like-for-like spectral benchmark under three crisis labellings, the adaptive- k attribution control, the lead-lag analysis, the disentangled large-network robustness check and the persistent-homology extension, the evidence supports a specific and bounded conclusion. TSI matches a properly sharpened spectral measure at detecting stress rather than beating it. It equals one, from a different construction, with the cleanest alarms of anything we tested, and it brings with it an attribution of each reading to a specific asset that needs no component count to be chosen and therefore cannot be chosen wrongly. That combination is what we would ask a practitioner to weigh, complementary to and not a replacement for established indices such as the Absorption Ratio.

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